

A Claim-Bounded Failure-Boundary Audit of an A10-Derived Hubble-Tension Scaffold

From Candidate Proxy to Physical-Bridge Non-Establishment

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Abstract

This manuscript reports a claim-bounded failure-boundary audit of an A10-derived Hubble-tension scaffold. The paper does not claim a solution to the Hubble tension, a replacement for Λ CDM, an observational discovery, or a real-data likelihood result. The audited chain was designed to determine whether A10-origin materials could be reduced to a form suitable for later cosmological comparison. The terminal result is negative but informative: a candidate proxy, an internal diagnostic curve, a contract-only bridge scaffold, and a bounded hypothesis scaffold were retained; however, no first-principles physical background-expansion equation $E(z)$ or $H(z)$ was established inside the audited artifacts. Consequently, the chain did not proceed to Λ CDM reference smoke, real-data likelihood testing, matched baseline comparison, MCMC, posterior comparison, or evidence evaluation. The principal contribution is therefore not a new cosmological model, but a documented failure-localization record: the unresolved boundary lies between candidate scaffold material and a physically typed background cosmology interface. This record is preserved as a methodological closeout and as input to a possible future rebuild of A10 Core as a structured-prior and strict-audit research operating procedure rather than as a validated physical theory.

Keywords: Hubble tension; failure-boundary audit; claim boundary; A10 methodology; cosmology scaffold; negative result; research governance.

1 Introduction

The Hubble tension motivates many proposals that attempt to modify the early or late expansion history of the Universe. Within the author’s A10 project, a Hubble-tension branch was previously explored as a structured candidate involving a localized early-time transfer, stability-gated selectivity language, and a high- H_0 target region. Earlier A10 Core documents deliberately limited the public claim: A10 Core was framed as a structured-prior methodology, not as a universal physical law or a domain-validated cosmological theory [1]. The same documents also stated that stronger cosmological claims would require independent reproduction, likelihood comparison, robustness testing, and expert review.

This paper records the subsequent audit rather than promoting the earlier scaffold. The audit asked a stricter question: after deconstruction, could the A10-derived candidate proxy be connected to a first-principles physical background expansion equation, such as $E(z)$ or $H(z) = H_0 E(z)$, strongly enough to justify a later Λ CDM reference smoke or real-data likelihood comparison? The answer in the audited artifacts was no. This is not a numerical defeat by data, because the audit stopped before real data and before likelihood evaluation. It is a structural negative result: the physical bridge required to enter cosmological model comparison was not established.

The purpose of this manuscript is to make that stop point explicit. It records what survived, what did not, and why the chain stopped. It also seals the public claim boundary so that the

archive cannot be misread as a Hubble-tension solution or as an observational model-selection result.

2 A10 Core context and audit object

The relevant A10 Core position is methodological. A10 Core v0.2.1 describes the framework as a structured-prior and claim-boundary methodology for constructing diagnostic branches, not as a single equation that overrides domain science [1]. Its operational sentence is: bold hypothesis, strict process, bounded claim. Under that rule, unusual hypotheses may enter an exploratory workflow, but only evidence-bounded claims may exit it.

The object audited here was not A10 Core as a general workflow. The audited object was narrower: an A10-derived Hubble-tension scaffold, including candidate proxy material, bridge-hypothesis attempts, and terminal handoff packages generated after a deconstruction workflow. The pre-audit seed dossier had already begun stripping A10-specific identity from the H0 branch, separating carry-forward candidates from deferred, ablated, or archived components [2]. Table 1 summarizes that stripping ledger.

Table 1: A10 element stripping ledger used as pre-audit source material.

ID	Status	Component
C01	Carry forward	Generic localized early-time transfer kernel near $z \approx 5000$.
C02	Carry forward	Quasi-vacuum to stiff-fluid reduced surrogate.
C03	Carry forward	Perturbation-level validation requirement.
C04	Carry forward	Residual-control objective under Λ CDM and benchmark comparators.
C05	Defer or ablate	Generic stability-gated activation; fixed $\chi \approx 137$ only as ablation coordinate.
C06	Defer or ablate	Spectral-Horndeski stability-trigger idea.
C07	Archive only	Root-geometric / E8 / ADE selectivity language.
C08	Defer or ablate	Late-time drag / S_8 companion sector.
C09	Defer or ablate	High- H_0 candidate behavior.

The audit therefore did not attempt to preserve A10 branding. It attempted to determine whether any neutral residue could become a physically typed cosmological seed. The answer was bounded: some scaffold material remained, but the physically required bridge remained absent.

3 Audit protocol

The protocol deliberately separated allowed documentation operations from forbidden scientific operations. Allowed operations included hash verification, extraction of candidate specifications, symbolic guard checks on toy grids, derivation-gap matrices, no-go gate records, and terminal handoff synthesis. Forbidden operations included real-data ingestion, parser or adapter execution for external datasets, likelihood or chi-square evaluation, matched Λ CDM baseline execution, MCMC, posterior comparison, evidence metrics, and public solution claims.

This design was intentional. If a physical $E(z)/H(z)$ bridge could not be established, then moving to real-data likelihood or Λ CDM comparison would only create a false appearance of rigor. The audit therefore used `D2_READY=False` as the decisive gate condition.

4 Deconstruction chain summary

Table 2 summarizes the compact terminal chain reported in this repository. Earlier fine-grained modes were consolidated into compound packages when their outcomes were expected documenta-

tion or guard outcomes. Unexpected results would have required separate re-inspection; none of the terminal packages authorized real-data comparison.

Table 2: Audit-chain summary.

Package	Role	Main result	D2 ready
Phase 1.5	A10 stripping and neutral seed dossier	A10-origin components were split into carry-forward, defer/ablate, and archive-only categories.	No
F2	Symbolic-contract guard smoke	A normalized symbolic scaffold passed finite/positive toy-grid guards only.	False
F3	Physical-status / derivation-gap matrix	The symbolic scaffold was separated from physical cosmology status.	False
F4–F7	Compound derivation-gap and D2 gate	D2 prerequisites remained unmet; no likelihood or comparison was executed.	False
G0–G3	First-principles-or-null decision	No first-principles physical $E(z)/H(z)$ derivation was found in the current artifacts.	False
H0–H3	Negative derivation close-out	Candidate scaffold was retained only as a bounded hypothesis scaffold.	False
I0–I3	Human-readable terminal dossier	Expert handoff and no-go gate were synthesized.	False
J0–J3	Terminal QA and public boundary	Public boundary was sealed; recommended next action was stop, restart, or expert handoff.	False

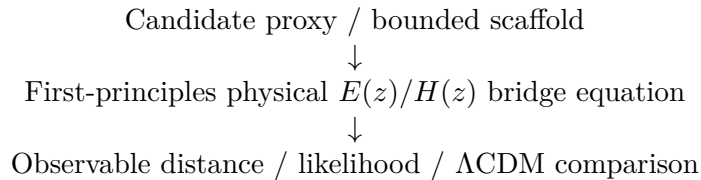
The key feature of this chain is the difference between a scaffold that is finite under toy guards and a physical cosmological model. The former is not sufficient for the latter. Passing a symbolic or contract-only guard means that the expression was well-behaved under a limited synthetic domain. It does not imply that the expression has a physical derivation, unit-consistent interpretation, observable-distance mapping, or likelihood-ready implementation.

5 Main result: physical bridge non-establishment

The terminal result can be stated in one sentence:

The A10-derived H0 scaffold retained bounded diagnostic material, but no first-principles physical $E(z)/H(z)$ bridge was established in the audited artifacts.

The main failure boundary is therefore not simply “the theory failed” in a vague sense. It is located at a specific interface:



The first line retained material. The second line was not established. The third line was therefore not entered. This is a structural negative result before observational comparison.

6 Retained material

The audit retained the following materials as bounded handoff artifacts:

- a candidate proxy;
- an internal diagnostic curve;
- a contract-only bridge scaffold;
- a distance-like integral guard, explicitly not an observable distance;
- bridge-hypothesis sandbox candidates;
- a failure-localization record;
- a human-readable expert handoff synthesis.

These items are potentially useful as diagnostic records or as inputs to a future model-building discussion. They are not sufficient to define a cosmological model.

7 Material not established

The following items were not established:

- a first-principles physical $E(z)$ equation;
- a first-principles physical $H(z) = H_0 E(z)$ equation;
- an observable cosmological distance model;
- a SN/BAO/CMB likelihood;
- a matched Λ CDM baseline comparison;
- MCMC, posterior comparison, or evidence evaluation;
- a Hubble-tension solution claim;
- a Λ CDM replacement claim.

This distinction is the central claim boundary of the repository.

8 Unauthorized operations remained zero

The terminal audit chain maintained the intended firewall. The forbidden operations listed in Table 3 were not executed. This does not make the scaffold physically successful; it makes the negative result clean. The archive does not mix a scaffold-level stop with unauthorized observational claims.

Table 3: Forbidden operation counters at terminal closeout.

Operation	Count
Real-data use or ingestion	0
Parser / adapter execution for external data	0
Real likelihood or chi-square evaluation	0
Matched Λ CDM baseline execution	0
MCMC execution	0
Posterior comparison	0
Evidence metric, Bayes factor, AIC, or BIC emission	0
Hubble-tension solution claim emission	0
Λ CDM replacement claim emission	0

9 Public claim boundary

The public claim supported by this archive is narrow:

An A10-derived H_0 scaffold was deconstructed and audited. Candidate proxy/scaffold material survived as bounded hypothesis material, but the physical $E(z)/H(z)$ bridge required for cosmological comparison was not established. Therefore no Λ CDM comparison, real-data likelihood, posterior, or evidence claim is made.

The archive does not support statements that A10 solved the Hubble tension, defeated Λ CDM, replaced Λ CDM, or obtained observational support from SN/BAO/CMB likelihoods. It also does not support the opposite overclaim that Λ CDM numerically defeated A10 in a matched likelihood contest, because such a contest was not executed.

10 Implications for A10 Core rebuild

The audit has an implication for A10 Core itself. If A10 Core is treated as a physical theory whose historical crown is the Hubble-tension branch, then this audit is damaging: the branch did not close as a physical background cosmology model. If A10 Core is instead treated as a strict research operating procedure, the result is more constructive. The workflow successfully stopped claim inflation, preserved the failure boundary, and demoted a tempting scaffold to bounded handoff material.

The rebuild implication is therefore clear. A10 Core should not use the cosmology branch as a success emblem. It may use the branch as a historical case study showing why strict audit and no-go ledgers are needed. In that sense, the cosmology branch becomes a record of self-audit rather than a crown.

Future A10-derived cosmology work, if any, should not begin by defending A10 identity. It should begin from a physically typed question: can any neutral residue be formulated as an effective-fluid deformation, modified background equation, or explicitly phenomenological diagnostic parameterization with unit consistency, observable mapping, and comparator discipline? If the answer is no, the cosmology route should remain closed.

11 Why this is not a data-level rejection

A common misunderstanding of this archive would be to read the result as if the A10-derived scaffold had been tested against data and had lost. That is not what happened. A data-level rejection would require a clearly defined physical model, a consistent mapping to observables, a matched baseline, a likelihood contract, and a comparison protocol. The present audit did not reach that level because the physical background interface was not established.

The negative result is therefore upstream of observational cosmology. It says that the audited artifacts did not contain enough physically typed structure to justify a comparison. In practical terms, this is a more basic stop than an unfavorable Bayes factor or a poor chi-square. It is the absence of a model-ready background equation.

This distinction matters for both humility and accuracy. The scaffold should not be advertised as having survived data, but neither should it be described as having been defeated by a data vector that was never loaded. Its status is: structurally incomplete for model comparison.

12 Relation to a possible neutral H_0 seed

The Phase 1.5 seed dossier proposed a neutralized future frame: a compact pre-recombination transfer kernel evaluated without A10 identity, without a privileged $\chi \approx 137$ assumption, and without high- H_0 alone as a success criterion [2]. The current audit strengthens that instruction.

If any future project restarts, it should not be called an A10 Hubble solution project. It should begin as a comparator-first residual-response audit.

A viable future seed would need at least four properties:

1. a unit-consistent physical or phenomenological definition of the background expansion modification;
2. an explicit mapping from that modification to observable distances or perturbative observables;
3. benchmarks against Λ CDM, ΔN_{eff} -like, early-dark-energy-like, no-gate, and generic-gate controls;
4. a stop rule that demotes the seed if it cannot pass perturbative, residual-cost, and prior-sensitivity tests.

The seed dossier already recommended that root-geometric, E8, and ADE selectivity language be archived rather than carried forward as physical cosmological evidence. The present audit makes that recommendation even stronger. Such language may remain historically useful, but it should not be part of any future physically typed H0 model unless independently rebuilt by domain specialists.

13 Decision matrix for next actions

Table 4 summarizes the appropriate next-action matrix. The table is included to prevent a common failure mode: treating a negative bridge result as an invitation to continue directly into a more complex pipeline. The audit recommends the opposite. Complexity should not be added until the missing interface is addressed.

Table 4: Next-action decision matrix.

Action	Condition	Status
Stop and archive	No physical $E(z)/H(z)$ bridge exists in the audited artifacts.	Recommended and sufficient.
Expert handoff	A reviewer is asked whether any scaffold can be reformulated as an effective-fluid, modified-background, or phenomenological model.	Allowed with boundary.
A10 Core rebuild	The failure log is used to demote cosmology from success emblem to self-audit case study.	Recommended.
Neutral H0 seed project	A new typed equation or phenomenological parameterization is constructed independently of A10 branding.	Possible but not started here.
D2 / Λ CDM reference smoke	A physical background interface, observable mapping, and comparator contract are established.	Not allowed yet.
Public solution claim	Any claim that the Hubble tension is solved or that Λ CDM is replaced.	Forbidden.

A Terminal decision matrix

The terminal J0–J3 package preserved the following matrix in human-readable form [3]. The table is reproduced here as a compact publication summary.

Table 5: Terminal decision matrix reproduced from the terminal closeout package.

Item	Decision	Comment
Candidate proxy	Retain	Internal diagnostic / bounded scaffold only.
Contract-only bridge scaffold	Retain	Not a physical equation.
Distance-like integral guard	Retain	Not an observable cosmological distance.
First-principles physical $E(z)$ derivation	Not found	Not established inside current artifacts.
Physical $H(z) = H_0 E(z)$ equation	Not established	Unit, derivation, and observable mapping remain unresolved.
D2 Λ CDM reference smoke	No-go	Preconditions unmet.
Real-data likelihood	No-go	Remained forbidden.
MCMC / posterior / evidence	No-go	Remained forbidden.
Public solution claim	Forbidden	Would violate the claim boundary.
Expert handoff	Allowed	Only as bounded negative/diagnostic handoff.

B No-go statement for reuse

For repository reuse, the shortest safe statement is:

This archive documents a claim-bounded failure-boundary audit. It retained candidate scaffold material but did not establish a first-principles physical $E(z)/H(z)$ bridge. Therefore it does not report a real-data likelihood, Λ CDM comparison, MCMC posterior, evidence metric, Hubble-tension solution, or Λ CDM replacement claim.

Any derivative document should preserve this statement or an equivalent boundary.

C Limitations

This manuscript is limited by design. It does not reproduce the full historical A10 cosmology chain, it does not load external data, and it does not independently run CLASS, Cobaya, or any full CMB/BAO/SN likelihood. The audit is therefore not an observational cosmology paper. It is a methodological and archival closeout paper. Its value lies in exact claim separation: scaffold retained; physical bridge not established; comparison not authorized.

D Conclusion

The A10-derived H_0 scaffold did not mature into a likelihood-ready cosmological model in the audited chain. The central missing piece was a first-principles physical $E(z)/H(z)$ bridge equation. Because that bridge was not established, the audit stopped before Λ CDM reference smoke, real-data likelihood, MCMC, posterior comparison, or evidence evaluation.

This is a negative result, but not an empty one. The archive preserves the exact failure boundary and the remaining bounded materials. It also demonstrates the stricter A10 posture in practice: bold hypotheses may enter, but unsupported claims do not exit. The correct final status is therefore not victory and not observational defeat, but claim-bounded failure-localization.

Repository and Data Availability

The repository package associated with this failure-boundary audit is available at:

<https://github.com/yokken0907/a10-derived-h0-scaffold-failure-boundary-audit>

The repository is provided as an audit archive and reproducibility aid. It is not an independent publication claim, and no repository release DOI or Zenodo DOI is assigned in this manuscript. The repository does not report a real-data likelihood, Λ CDM comparison, MCMC posterior, evidence metric, Hubble-tension solution, or Λ CDM replacement claim. Any future repository release or archival DOI should preserve the same claim boundary and should not be treated as a substitute for the present preprint record.

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